

Homework 9

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$$1. f_{XY}(x,y) = \begin{cases} c(x+y) & 0 < x < 1, 0 < y < 1-x \\ 0 & \text{otherwise} \end{cases}$$

$$a) \int_0^1 \int_0^{1-x} c(x+y) dy dx = \int_0^1 c x(1-x) + \frac{1}{2} c (1-x)^2 dx$$

$$= \int_0^1 -\frac{c}{2} x^2 + \frac{c}{2} dx = -\frac{c}{6} + \frac{c}{2} = 1$$

$$b) f_X(x) = \int_0^{1-x} f_{XY}(x,y) dy$$

$$= \int_0^{1-x} 3(x+y) dy = 3x(1-x) + \frac{3}{2}(1-x)^2 = 3x - 3x^2 + \frac{3}{2}(1-2x+x^2)$$

$$= -\frac{3}{2}x^2 + \frac{3}{2}$$

$$f_Y(y) = \int_0^{1-y} f_{XY}(x,y) dx$$

$$= \int_0^{1-y} 3(x+y) dx = -\frac{3}{2}y^2 + \frac{3}{2}$$

$$c) E[X] = \int_0^1 x f_X(x) dx = \frac{3}{2} \int_0^1 x(1-x^2) dx = \frac{3}{2} \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{3}{2} \times \frac{1}{4} = 3/8$$

$$V[X] = E[X^2] - E[X]^2$$

$$E[X^2] = \int_0^1 x^2 f_X(x) dx = \frac{3}{2} \int_0^1 x^2 - x^4 dx = \frac{3}{2} \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{3}{2} \cdot \frac{2}{15} = \frac{1}{5}$$

$$V[X] = \frac{1}{5} - \frac{9}{64} = \frac{64-45}{320} = 19/320$$

$$E[Y] = E[X] = 3/8 \quad V[Y] = V[X] = 19/320$$

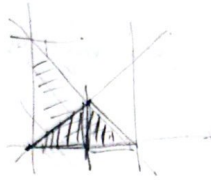
d) $P(X > Y)$

$$\int_0^1$$

$$P(X > Y) = P(Y > X)$$

$$P(X > Y) + P(Y > X) = 1$$

$$\therefore P(X > Y) = 1/2$$



$$P(X > Y) = \int_0^{1/2} \int_0^x c(x+y) dy dx + \int_{1/2}^1 \int_0^{1-x} c(x+y) dy dx$$

$$= \int_0^{1/2} 3x^2 + \frac{3}{2}x^2 dx + \int_{1/2}^1 3(x-x^2) + \frac{3}{2}(1-x)^2 dx$$

$$= \frac{1}{8} + \frac{1}{16} + \int_{1/2}^1 \left(\frac{3}{2} - \frac{3}{2}x^2 \right) dx$$

$$= \frac{3}{16} + \frac{3}{2} \cdot \frac{1}{2} - \frac{1}{2}x^3 \Big|_{1/2}^1 = \frac{3}{16} + \frac{3}{4} - \frac{1}{2} + \frac{1}{16} = 1/2$$

2. $x \in [0, 1]$

$$S(x) = x^r$$

$$X \sim \text{Beta}(a, b)$$

$$f_X(x) = \frac{1}{\beta(a, b)} x^{a-1} (1-x)^{b-1} \quad 0 \leq x \leq 1$$

$$\beta(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

$$a) f_{X|S}(x) = \frac{P(S|X)P(X)f_X(x)}{\sum_x P(S|X)f_X(x) = P(S)} = \frac{x^r \frac{1}{\beta(a, b)} x^{a-1} (1-x)^{b-1}}{\int_0^1 x^r \frac{1}{\beta(a, b)} x^{a-1} (1-x)^{b-1} dx} = \frac{x^{a+r-1} (1-x)^{b-1}}{\beta(a, b) \int_0^1 x^{a+r-1} (1-x)^{b-1} dx}$$

$$= \frac{x^{a+r-1} (1-x)^{b-1}}{\beta(a+r, b)}$$

$$\therefore X|S \sim \text{Beta}(a+r, b)$$

$$b) E[X] = \int_0^1 x f_X(x) dx = \frac{1}{\beta(a, b)} \int_0^1 x^{a+1-1} (1-x)^{b-1} dx = \frac{\beta(a+1, b)}{\beta(a, b)} = \frac{\frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+b+1)}}{\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}}$$

$$= \frac{\Gamma(a+b) \Gamma(a+1)}{\Gamma(a+b+1) \Gamma(a)} = \frac{a}{a+b}$$

$$E[X|S] = \frac{\Gamma(a+b+r) \Gamma(a+r+1)}{\Gamma(a+b+r+1) \Gamma(a+r)} = \frac{a+r}{a+b+r}$$

3. $R \sim U(0,1)$

a) given $R=r$, $P_{X|R=r}(x)$

$X|R=r \sim \text{Bin}(M, r)$

$$\therefore P(X=x|R=r) = \binom{M}{x} r^x (1-r)^{M-x}$$

$$\frac{1}{\beta(1,1)} x^0 (1-x)^0 = \frac{\Gamma(2)}{\Gamma(1)\Gamma(1)} = 1$$

b) $E[X] = E[E[X|R]] = \int_0^1 E[X|R=r] f_R(r) dr$

$$P(X=x) = \int_0^1 \frac{M!}{x!(M-x)!} r^x (1-r)^{M-x} dr = \frac{M!}{x!(M-x)!} \beta(x+1, M-x+1)$$

$$= \frac{M!}{x!(M-x)!} \cdot \frac{x! (M-x)!}{(M+1)!} = \frac{1}{M+1}$$

c) $V[X] = E[X^2] - E[X]^2$ $\therefore E[X] = \sum_{x=0}^M \frac{x}{M+1} = \frac{1}{M+1} \cdot \frac{M(M+1)}{2} = \frac{M}{2}$

$$V[X] = E[X^2] - E[X]^2$$

$$E[X^2] = \sum_{x=0}^M \frac{x^2}{M+1} = \frac{1}{M+1} \cdot \frac{M(M+1)(2M+1)}{6} = \frac{M(2M+1)}{6}$$

$$\therefore V[X] = \frac{2M^2+M}{6} - \frac{M^2}{4} = \frac{1}{12}M^2 + \frac{1}{8}M$$

c) $R \sim \text{Beta}(a,b)$

$$E[X] = E[E[X|R]] = \int_0^1 E[X|R=r] f_R(r) dr = M \int_0^1 \frac{1}{\beta(a,b)} r \cdot r^{a-1} (1-r)^{b-1} dr$$

$$= M \frac{\beta(a+1,b)}{\beta(a,b)} = M \cdot \frac{\frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+b+1)}}{\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}} = M \cdot \frac{\Gamma(a+b)}{\Gamma(a+b+1)\Gamma(a)} = M \cdot \frac{a}{a+b}$$

$$V[X] = E[V(X|R)] + V[E[X|R]]$$

$$= E[MR(1-R)] + V[MR] = M(E[R] - E[R^2]) + M^2 V[R]$$

$$= M(E[R] - E[R^2]) + M^2(E[R^2] - E[R]^2)$$

$$= M \left(\frac{a(a+b+1) - a(a+1)}{(a+b)(a+b+1)} \right) + M^2 \left(\frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b)^2(a+b+1)} \right)$$

$$= M \left(\frac{ab}{(a+b)(a+b+1)} \right) + M^2 \left(\frac{ab}{(a+b)^2(a+b+1)} \right) = \frac{Mab(a+b) + M^2ab}{(a+b)^2(a+b+1)}$$

$$E[R] = \frac{\beta(a+1,b)}{\beta(a,b)} = \frac{a}{a+b}$$

$$E[R^2] = \frac{\beta(a+2,b)}{\beta(a,b)} = \frac{a(a+1)}{(a+b)(a+b+1)}$$

$$\begin{aligned} & (a^2+a)(a+b) \\ & a^3 + a^2b + a^2 + ab - a^3 - a^2b - a^2 \end{aligned}$$

4. Gubner, Chapter 4, Problem 28

$$Z \sim \mathcal{N}(0, 1^2) \quad Y = Z + n \sim \mathcal{N}(n, 1^2)$$

$$\mathbb{E}[Z^{2m+1}] = 0 \quad \mathbb{E}[Z^{2m}] = (2m-1)!!$$

$$\mathbb{E}[(Y-n)^4] = 3 \cdot 1^2 = 3 \quad -4n\mathbb{E}[Y^3] - 4n^2\mathbb{E}[Y]$$

$$\mathbb{E}[Y^4 - 2nY^3 + n^2Y^2 - 2n^3Y + n^4] = \mathbb{E}[Y^4] + 6n^2\mathbb{E}[Y^2] + n^4 = 3$$

$$\mathbb{E}[Y^2] = V[Y] + \mathbb{E}[Y]^2 = 1 + n^2$$

$$\mathbb{E}[Y^4] = 3 - n^4 - 6n^2(1 + n^2) + 4n\mathbb{E}[Y^3] + 4n^2\mathbb{E}[Y]$$

$$\mathbb{E}[(Z+n)^3] = \mathbb{E}[Z^3] + 3n\mathbb{E}[Z^2] + 3n^2\mathbb{E}[Z] + n^3$$

$$\mathbb{E}[Y^4] = \mathbb{E}[(Z+n)^4] = \mathbb{E}[(Z^2 + 2nZ + n^2)(Z^2 + 2nZ + n^2)]$$

$$= \mathbb{E}[Z^4 + 2nZ^3 + n^2Z^2 + 2nZ^3 + 4n^2Z^2 + 2n^3Z + n^2Z^2 + 2n^3Z + n^4]$$

$$= \mathbb{E}[Z^4] + 6n^2\mathbb{E}[Z^2] + n^4$$

$$= 3 \cdot 1 + 6n^2 \cdot 1 + n^4 = n^4 + 6n^2 + 3$$

5. Gubner, Chapter 4, Problem 35

$$X_i \sim \text{Exp}(\lambda)$$

$$\mathbb{E}[X_i] = 1/\lambda = 20$$

$$\lambda = 1/20$$

$$P(X_i \geq 25) = \int_{x=25}^{\infty} \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_{25}^{\infty} = e^{-25\lambda}$$

$$= e^{-5/4}$$

$$P(Y) = 1 - \prod_{i=1}^5 P(X_i < 25) = 1 - (1 - e^{-5/4})^5$$

6. Gubner Chapter 4, Problem 52

$$Z = X + Y$$

$$X \sim \exp(1), Y \sim \text{Laplace}(1)$$

Find $\text{Cov}(X, Z)$, $\text{Var}(Z)$

$$\text{Cov}(X, Z) = E[XZ] - E[X]E[Z]$$

$$= E[X^2 + XY] - E[X]E[X] - E[X]E[Y]$$

$$= E[X^2] - E[X]^2 + E[XY] - E[X]E[Y]$$

$$= \text{Var}(X) + \text{Cov}(X, Y) = 1$$

$$\text{Var}(Z) = \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) = 1 + 2 = 3$$

$$f_X(x) = \lambda e^{-\lambda x}, x \geq 0$$

$$E[X] = 1/\lambda = 1, E[X^2] = 2/\lambda^2 = 2$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = 2 - 1 = 1$$

$$f_Y(y) = \frac{\lambda}{2} e^{-\lambda|y|}, y \in \mathbb{R}$$

$$E[Y] = \int_{-\infty}^0 \frac{1}{2} y e^y dy + \int_0^{\infty} \frac{1}{2} y e^{-y} dy = 0$$

$$\text{Var}(Y) = E[Y^2] - E[Y]^2$$

$$\int_{-\infty}^0 \frac{1}{2} y^2 e^y dy + \int_0^{\infty} \frac{1}{2} y^2 e^{-y} dy$$

$$= \int_0^{\infty} y^2 e^{-y} dy = \Gamma(3) = 2$$

$$\therefore \text{Var}(Y) = 2 - 0 = 2$$

7. Gubner Chapter 5, Problem 6

$$Y = 1 - X \text{ show } f_Y(y) = f_X(1 - y)$$

$$\int_a^b f_X(x) dx = \int_{1-a}^{1-b} f_Y(1-x) (-dy) = \int_{1-b}^{1-a} f_Y(1-x) dy$$

$$\int_a^b f_Y(y) dy = \int_{1-a}^{1-b} f_X(1-y) (-dx) = \int_{1-b}^{1-a} f_X(1-y) dx$$

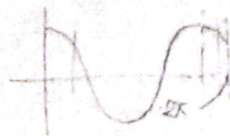
$$\text{if } X \sim \text{unif}(0,1) \quad f_X(x) = 1 \quad 0 \leq x \leq 1$$

$$\Rightarrow f_Y(1-x) = 1 \quad 0 \leq x \leq 1$$

$$= f_Y(y) = 1 \quad 0 \leq y \leq 1$$

$$\therefore Y \sim \text{unif}(0,1)$$

8 Gubner, Chapter 7, Problem 39



a) $Y|X=x \sim \text{Unif}(x-\pi, x+\pi)$

$$\mathbb{E}[\cos(X+Y)] = \mathbb{E}[\mathbb{E}[\cos(X+Y)|X]]$$

$$= \int_{-\infty}^{\infty} \mathbb{E}[\cos(X+Y)|X=x] f_X(x) dx = \int_{-\infty}^{\infty} \mathbb{E}[\cos(Y+x|X=x)] f_X(x) dx$$

$$Y+x|X=x \sim \text{Unif}(2x-\pi, 2x+\pi)$$

$$\mathbb{E}[\cos(Y+x|X=x)] = \int_{x-\pi}^{x+\pi} \cos(x) dx = 0$$

$$\therefore \mathbb{E}[\cos(X+Y)] = 0$$

b) $X \sim \text{Unif}(1,2)$ $Y|X=x \sim \text{Exp}(x)$

$$f_{Y|X}(y|x) = x e^{-xy} = \frac{f_{XY}(x,y)}{f_X(x)}$$

$$f_{XY}(x,y) = x e^{-xy} \cdot 1 \quad 1 \leq x \leq 2$$

$$f_Y(y) = \int_1^2 f_{XY}(x,y) dx = \int_1^2 x e^{-xy} dx = -\left(\frac{2}{y} + \frac{1}{y^2}\right) e^{-2y} + \left(\frac{1}{y} + \frac{1}{y^2}\right) e^{-y} \quad y > 0$$

$$P(Y > y) = \int_y^{\infty} f_Y(y) dy = \frac{e^{-y} - e^{-2y}}{y}$$

c) $X \sim \text{Unif}(3,7)$ $Y|X=x \sim \mathcal{N}(0, x^2)$

$$\mathbb{E}[x e^Y] = \mathbb{E}[\mathbb{E}[x e^Y | X]] = \int_3^7 x \mathbb{E}[e^Y | X=x] \frac{1}{4} dx$$

$$\mathbb{E}[e^Y | X=x] = \int_{-\infty}^{\infty} e^y f_{Y|X}(y|x) dy = \int_{-\infty}^{\infty} e^y \frac{1}{\sqrt{2\pi} x} e^{-\frac{y^2}{2x^2}} dy = e^{x^2/2}$$

$$\frac{1}{4} \int_3^7 x e^{x^2/2} dx = \frac{1}{4} [e^{x^2/2}]_3^7 = \frac{1}{4} (e^{49/2} - e^{9/2})$$

d) $X \sim \text{Unif}(1,2)$ $Y|X=x \sim \mathcal{N}(0, 1/x)$

$$\mathbb{E}[\cos(XY)] = \int_1^2 \mathbb{E}[\cos(XY) | X=x] f_X(x) dx = \int_1^2 \mathbb{E}[\cos(XY) | X=x] dx$$

$$\mathbb{E}[\cos(XY) | X=x] = \int_{-\infty}^{\infty} \cos(xy) f_{Y|X}(y|x) dy = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} x} \exp\left(-\frac{y^2}{2x}\right) \cos(xy) dy$$

$$= \int_{-\infty}^{\infty} \frac{x}{\sqrt{2\pi}} \exp\left(-\frac{xy^2}{2}\right) \cos(xy) dy = \exp(-x/2)$$

$$\therefore \mathbb{E}[\cos(XY)] = \int_1^2 e^{-x/2} dx = -2e^{-x/2} \Big|_1^2 = -2e^{-1} + 2e^{-1/2}$$